

Author: As to the most frequent functions such as power, exponential, logarithmic, trigonometric, and inverse trigonometric, they are continuous at all points of the natural domains of the corresponding analytical expressions. The same can be said about composite functions obtained from the above elementary functions. The continuity of all these functions is proved in the more advanced courses of calculus. We limit ourselves to a mere stating of the fact.

Dialogue Seven: More on the Limit of Function

Reader: Comparing the definition of the limit of a function at a point with the definition of the limit of a numerical sequence, I come to the conclusion that these two limits are of different nature.

Author: And I understand why. In fact, I did emphasize the difference myself in the previous dialogue, pointing out, as you probably remember, that a sequence is a function defined on a set of integers, while the functions we are discussing at the moment are defined over intervals. I doubt, however, that you are justified in speaking about the difference in the nature of the limit of function and that of sequence. In the final analysis (and this is essential) the limit of a function at a point may be defined on the basis of the limit of a numerical sequence.

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Reader: This is very interesting.

Author: Let us forget, for the time being, about the definition of the limit of function given in the previous dialogue. Consider a new definition. We shall consider, as before, a function $f(x)$ defined over an interval, and a point $x=a$ either taken within the interval or coinciding with its end.

Author: The two are equivalent.

Reader: But in form they are quite different!

Author: We can prove their equivalence. To begin with, let the definition using a δ -neighbourhood of point a be called "definition 1", and the definition using numerical sequences, "definition 2".